Information content of option implied eigenportfolios and their variance risk premium sensitivities in G10 currencies*

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Abstract

This paper reveals a novel way of constructing the option implied RX and HML_{FX} risk factors by utilizing the full cross-section of the 45 G10 cross-rate options. The option implied carry factors, $IEP_{\tau}^{2,ZC}$, are highly profitable strategies that surprisingly earn both positive carry and positive spot price contribution and stochastically dominate the traditional HML_{FX}. Moreover, the changes of 6m-1m variance risk premium slope of implied dollar factor, IEP_{τ}^{1}, have predictive power on both $IEP_{\tau}^{2,ZC}$'s and HML_{FX} 's returns. We conclude the implied eigenportfolios to contain information to both improve the profitability of carry strategies via allocation and via predicting the strategy related risks.

JEL classification: G10; G15; G30; G32

foreign exchange, implied volatility, portfolio distribution

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1 Introduction

This paper provides a novel way of revealing the option embedded currency risk factor allocations for option implied RX and HML_{FX} in G10 currencies. These forward looking market implied eigenportfolios create highly profitable carry strategies, $IEP_{\tau}^{2,ZC}$, that stochastically dominate HML_{FX} and earn both high carry and surprisingly high spot price contribution as well. We also find new empirical evidence that the expected carry retruns of both $IEP_{\tau}^{2,ZC}$ and HML_{FX} are strongly related to the changes of the 6m-1m slope of variance risk premium term structure (VTS) of the implied dollar factor, IEP_{τ}^{1}. These VRP's are measured as a τ month backward looking realized variance minus τ month forward looking implied variance on market implied dollar portfolios. We find that the decrease in the 6m-1m VTS leads to decrease in carry strategies profitability in the coming month.

We execute our empirical study against the background of consistent violations in the uncovered interest parity (UIP) and the plethora of option related carry trade risk studies. Despite the wast amount of option related research, surprisingly no papers utilize the option information to find the forward looking carry trade allocation. Due to limited availability of the option data for the full cross-section, we focus on the after crisis period on G10/USD currencies from 09/2008 to 08/2016.

We utilize the full cross-section of 45 volatility currency cross rates to back out time- series of implied covariance matrices for four different maturities ($\tau \in [1m, 3m, 6m, 12m]$). These matrices are decomposed into matrices of implied eigenportfolios (IEP) and their corresponding eigenvalues, using principal component analysis (PCA). As in the risk factor literature (see for example Lustig et al. (2011)) we focus on the first two, which have clear interpretations in the currency market, the dollar (RX) and the carry (HML_{FX}) factors. The first normalized eigenportfolios, $IEP_{t,\tau}^{1}$, represents forward looking time-varying option implied dollar factor allocation, whereas the second normalized eigenportfolios represent the respective allocation for implied carry factor $IEP_{t,\tau}^{2}$. In order to achieve the orthogonality between $IEP_{t,\tau}^{1}$ and $IEP_{t,\tau}^{2}$, PCA allocates extra leverage (varies between -80% to 80% in our sample) on either short or the long leg. We therefore also construct a set of their zero-cost counterparty, $IEP_{t,\tau}^{2,zc}$,

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having the same allocation within short and long legs, but forcing the dollar amount invested in both legs to be equal. This obviously effects the orthogonality, but resembles the traditional HML_{FX} more. The difference of these two we call $Orth_\tau = IEP^{2,zc}_{t,\tau} - IEP^{2}_{t,\tau}$. We utilize a nonparametric stochastic dominance (SD) test to examine the efficiency implications of carry trades. The main logic of the SD test is simple: if the distributions of implied carry factors, $IEP^{2,zc}_{t,\tau}$ or $IEP^{2}_{t,\tau}$, returns are preferred by all expected utility maximizers over HML_{FX} (an efficiently priced risky asset), then the option markets contains allocation information that is not risk based compensation, but contains some superior information to increse the profitability. Since SD test is nonparametric, the SD based statistical inferences are independent on any asset pricing model or return normality assumptions. We find the option implied carry strategy, $IEP^{2,zc}_{t,\tau}$, to be highly profitable and to stochastically dominate (first and second order) the traditional HML_{FX} strategy. There exist a large body of literature proving the profitability of HML_{FX} to exist solely due to the positive yield differential returns, whereas spot price seem to have no contributions in the long run. Surprisingly the option implied carry strategy, $IEP^{2,zc}_{t,\tau}$, earns equally high interest rate differential contribution as HML_{FX}, but it also allocates wealth to currencies that have a high positive spot price contribution to the strategy as well. This spot price contribution seems to be driven mainly due to elimination of the extra leverage, $Orth_\tau = IEP^{2,zc}_{t,\tau} - IEP^{2}_{t,\tau}$.

Moreover, with the weight vectors for the option implied dollar factor, $IEP^{1}_{t,\tau}$, and the implied covariance matrices available, we calculate the term structure of VRP's. We regress the 6m-1m VTS against $IEP^{2,zc}_{t,\tau}$, $IEP^{2}_{t,\tau}$, $Orth_\tau$ and HML_{FX} excess returns and find the R^2's to vary between 1% and 12%. The predictability being high for the $IEP^{2,zc}_{t,\tau}$, HML_{FX} and $Orth_\tau$. This high predictive power of 6m-1m VTS of the $IEP^{1}_{t,\tau}$ is used as an indicator variable to take take either long or the short postion on the carry strategies, which clearly improves the profitability of the strategies.

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2 Literature

A currency carry trade is a trading strategy, which exploits the well documented forward premium puzzle (Bansal (1997), Baillie and Bollerslev (2000), Bansal and Dahlquist (2000), Backus et al. (2001), and Eichenbaum et al. (2007) Burnside et al. (2009), by investing in the high yielding currencies and funding it with low yielding currencies.

A rich empirical literature proves the long term profitability of the carry trades, but the profitability remains a puzzle unless the strategy returns can be shown to be a compensation for bearing risk. The literature is lacking consensus on the cause of the carry return profitability. Where as research by Lustig and Verdelhan (2007) , Lustig et al. (2014), Verdelhan (2015), and Colacito et al. (2015) link the cross-sectional evidence of carry trade profitability to fundamental risk factors (consumption, dollar, carry-trade, long-run growth), Burnside et al. (2006, 2008) argue the payoffs to carry trades to be uncorrelated with standard risk factors such as stock returns and consumption growth. Villanueva (2007) supports the puzzle argument by stating the carry trade Sharpe ratios to be too high to reflect risk compensation. Also Fong (2010) show yen carry trades superiority over S&P500 returns and uses the result as an argument for the puzzle. De Santis and Fornari (2008) present weak evidence that high yielding currencies have higher loadings on consumption growth risk.

A plethora of papers have also suggested the carry investors to require high returns on average as a compensation for crash risk. The merit of this hypothesis is justified by the magnitude and probability of large moves in currency markets. Vast efforts have been taken to comprehend the nature of these peso effects in carry trade returns (Brunnermeier et al. (2008), Jurek (2014), Dupuy (2015), Farhi et al. (2015), Daniel et al. (2016), and Chernov et al. (2018a)). The option market information has played a crucial role in understanding the carry trade related compensation of crash risks, but the research has mainly focused on volatility surfaces of individual currency pairs. These papers utilize currency options to run hedged carry strategies still yield significant profits, which suggest that crash risk may only partially be explain carry trade profitability.

More recent literature has focused on the predictive power of the Variance Risk Premium

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(VRP) in explaining the time variation in currency returns. Della Corte et al. (2011) finds that the VRP in currencies to be positive, time-varying, and predictable. Also Menkhoff et al. (2012) report the finding that global forex volatility risk is priced in currency markets (see alsoBakshi and Panayotov (2013)). Chernov et al. (2018b) report that jump risk in currency variance may be priced in currency markets but is unrelated to interest rates or macroeconomic news. Where as Farhi et al. (2009), Jurek (2014) and Brunnermeier et al. (2008) relate the high currency volatility to the desire of agents to hedge tale events in exchange rate movements. Finally Londono and Zhou (2017a) studies the predictive power of currency and equity VRP's in the cross-section of 22 X/USD crosses. Authors create a new equally-weighted VRP -index for currencies and concentrate on the predictive power of 6 month option tenor. They find the constructed index to have nonredundant and significant predictive power for the appreciation rates of 22 X/USD crosses. The risk factor VRP part of this paper is related to their work. Instead of using equally weighted index of VRP's, we use the true implied covariance matrix and the option implied weight vectors for the implied dollar factor to construct our VRP measure. In this way the market's expected covariance structure and the risk factor allocation are also taken into account. Instead of focusing just on the 6 month tenor, we focus on the 6m-1m VRP slope. An emerging literature of Dew-Becker et al. (2017), Eraker and Wu (2017), and Johnson (2016) document the downward sloping shape of the volatility risk premia term structure in equity markets, and Della Corte et al. (2016) report the same phenomenan also to exist in FX markets. A vast literature of VRP predicting power in different asset classes exists, but only a few studies concentrate on the VRP term structure. Bormetti et al. (2016) demonstrate the relevance of the whole VRP term structure in explaining equity prices, and establish a theoretical relation between the equity risk premium, VRP and the slope of VRP term structure with one-component GARCH model. As the term structure contains information about the future dynamics of VRP over different time horizons, they show that the shape of the term structure (summarized by the slope) has strong predictive power on future stock-index returns.

The aim of this paper is to extend the literature in two ways. First, it contributes to the

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option and risk factor literature by being a first paper to reveal the option implied dollar (RX) and carry HML_{FX} factors, and to examine whether these risk adjusted market implied carry trade allocations contains some superior information to increase the carry trade profitability. Second, the paper extends the VRP literature by examining the predicting power of the VRP term structure slope of these new option implied risk factors. There exists a very limited literature on VRP slope's information contents and no literature on risk factor VRP's.

3 Data

The results in this paper are based on two following datasets. The first data set is from Bloomberg and is used to form the currency risk return seriesses for the risk factor strategies. It contains 10 countries' (USD, EUR, JPY, GBP, CHF, AUD, NZD, CAD,NOK and SEK) 1 month Eurocurrency rates, which are the interest rates at which banks are willing to borrow and lend foreign currency deposits, and constitute the effective interest rates at which a carry trade investor would be able to borrow and lend. Data is month end closing prices and spans from 09/2008 to 08/2016. From the same data period, month end closing prices for nine G10/USD foreign exchange spot and forward rates are also collected. The second data set, the foreign exchange (FX) option dataset, is comprised of month end implied volatility quotes of ATM volatilities for 45 volatility crosses in G10 space. The FX options are European and give their owners the right to buy or sell a foreign currency at a pre-specified exchange rate measured in U.S. dollars per unit of foreign currency. FX option prices are quoted in terms of their Garman and Kohlhagen (1983) implied volatilities with their delta convention (spot delta or premium adjusted spot delta) depending on the currency-cross quoted and the volatility prices are quoted via the traditional market quotes 10d and 25d risk reversals and butterflies and ATM vol. The data on exchange rate options was obtained from Bloomberg and covers the period from 09/2008 to 08/2016.

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4 Methodology

This section introduces a novel way of utilizing the information contents of the G10/USD implied covariance matrix to reveal the time varying option implied risk factor allocations for RX and HML_{FX}. We employ principal component analysis (PCA) on the option implied covariance matrix and decompose it into orthogonal eigenportfolios (EP). This risk factors construction process is controlled by forcing the random sign of the first and second eigenvector to mimic the nature of realized risk factors. I.e. the first EP (IEP_τ^1) is the RX factor assigning mainly positive loadings on its elements and the second EP(IEP_τ^2) is the HML_{FX} factor, always yielding positive carry by construction. This way a set of orthogonal risk factors are produced. Unlike the traditional HML_{FX} factor, extra leverage is assigned on either long or short leg of the portfolio, which is why we also produce a set of zero-cost carry factors ($IEP_\tau^{2,ZC}$) by removing the extra leverage. The realized 1 month holding period returns are calculated and their superiority to the traditional carry returns are examined with stochastic dominance and Sharpe ratio tests. Finally, we utilize the EP weight vectors to calculate variance risk premium term structure for the implied dollar factor IEP_τ^1 and we examine its predictive power on IEP_τ^2, $IEP_\tau^{2,ZC}$ and HML_{FX} by using it as a timing tool.

4.1 Forming of the RX and HML_{FX} strategies

First we construct the weight vectors for the common risk factors, as presented in literature, so that the dollar factor weight vector, $\boldsymbol{W_{RX,t}}$, in every point in time is a $1xN$ vector of equal $1/9$ weights and the carry factor weightvector $\boldsymbol{W_{HML,t}}$ has $1/3$ weight on three highest yielding G10/USD currency pairs and $-1/3$ on the three lowest yielding G10/USD currency crosses. We follow much of the recent literature (Lustig and Verdelhan (2007), Lustig et al. (2011)) and sort the G10/USD crosses at the end of each month into 3 portfolios based on their forward discounts $log(F_t/S_t)$. All three portfolios are rebalanced at the end of each month, Portfolio 1 consisting the lowest forward premium currencies and portfolio 3 consisting of the highest forward premium currencies respectively. After one months holding period the monthly excess

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return for holding foreign currency i is calculated as

$$rx_{t+1} = i_t^k - i_t - \Delta s_{t+1}^k \approx f_t^k - s_{t+1}^k \tag{1}$$

Ranked currency specific log excess returns are divided into three equally weighted portfolios by averaging of the rx_{t+1}^k belonging to particular portfolio.

4.2 Calibration of the implied spot correlations

To obtain the implied covariance matrix $\boldsymbol{\Sigma_{t,\tau}}$ for G10/USD crosses, we first back out the implied correlation matrix, $\boldsymbol{\rho} t, \tau$, where $\tau \in [1, 3, 6, 12]$, utilizing ATM implied volatilities $\boldsymbol{\sigma}_{i,t,\tau}$ on 45 different currency crosses. As the so called currency triangle, the natural no arbitrage condition between spot prices (i.e. if EURUSD=g and USDJPY=h, EURJPY=g*h), has to exist, also the FX volatility has to be directly determined by the volatilites of the other two spot rate processes, together with the correlation between Brownian motions. I.e. $\sigma_{i,j}^2 = \sigma_i^2 + \sigma_j^2 2\rho_{i,j}\sigma_i\sigma_j$. This means that implied correlations can be obtained from implied volatility quotes for the three FX spot rates comprising the currency triangle

$$\rho^{i*j} = \frac{[\sigma^{i*j}]^2 - [\sigma^i]^2 - [\sigma^j]^2}{2\sigma^i\sigma^j} \tag{2}$$

$$\rho^{i/j} = \frac{[\sigma^{i/j}]^2 - [\sigma^i]^2 - [\sigma^j]^2}{2\sigma^i\sigma^j} \tag{3}$$

Note that three correlations can be found depending on which of the three vertices of the currency triangle is regarded as the common base currency. We could attempt to estimate the correlation between EURUSD and EURJPY, between EURUSD and USDJPY, or between EURJPY and USDJPY. The extension of this to multiple spot rates is relatively straightforward, assuming that a common base currency can be found, e.g. USDJPY, AUDJPY, EURJPY.

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4.3 Principal Component Analysis(PCA) as a risk factor construction tool

Consider the TxN return matrix vector of N assets in the portfolio:

$$\boldsymbol{R} = (\boldsymbol{R_i}...\boldsymbol{R_N}) \tag{4}$$

where every $\boldsymbol{R_i}$ is a $Tx1$ vector of asset returns. Principal component analysis is based on the spectral decomposition of a covariance matrix $\boldsymbol{\Sigma_{t,\tau}} = T^{-1}\boldsymbol{(R-\mu)'(R-\mu)}$ and the covariance matrix of $\boldsymbol{\Sigma}$ is defined as

$$\boldsymbol{\Sigma}_{t,\tau} = \boldsymbol{\sigma}_{t,\tau}\boldsymbol{\rho}_{t,\tau}\boldsymbol{\sigma}_{t,\tau} \tag{5}$$

Where $\boldsymbol{\sigma}$ is a NxN diagonal matrix of (implied) volatilities of N assets returns and $\boldsymbol{\rho}$ is a NxN (implied) correlation matrix of N assets returns. Next the spectral decomposition of a covariance matrix is calculated:

$$\boldsymbol{\Sigma_t} = \boldsymbol{W\lambda W'} \tag{6}$$

Where $\boldsymbol{\lambda}$ is a diagonal matrix of eigenvalues of the $\boldsymbol{\Sigma}$, sorted in descending order so that the eigenvalues are $\boldsymbol{\lambda_1} \geq \boldsymbol{\lambda_2} \geq ...\boldsymbol{\lambda_N} \geq 0$ and

$$\boldsymbol{W} = (\boldsymbol{W_i}...\boldsymbol{W_N}) \tag{7}$$

$$\boldsymbol{W_i} = (\boldsymbol{w_{i,1}}...\boldsymbol{w_{i,N}})' \tag{8}$$

$\boldsymbol{W}$ is a NxN eigenmatrix, where i^{th} column contains an eigenvector corresponding to eigenvalue $\boldsymbol{\lambda_i}$. These N eigenvectors are actually $Nx1$ weight vectors to form N orthogonal statistical factors, each explaining the individual dimension of risk.

$$\boldsymbol{F} = \boldsymbol{W^T(R-\mu)} \tag{9}$$

$$\mathbb{E}\boldsymbol{(FF^T)} = \boldsymbol{\lambda} \tag{10}$$

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$$\boldsymbol{W}\boldsymbol{W}^{\boldsymbol{T}} = \boldsymbol{I} \tag{11}$$

So the time-serieses of the factors $\boldsymbol{F}$ can be recovered using the eigenmatrix and the original return serieses. Nice feature of these factors is also, that their covariance matrix is the $\boldsymbol{\lambda}$, where $\boldsymbol{\lambda_i}(diag(\boldsymbol{\lambda})\mathbf{1}')^{-1}$ illustartes the total variance explained by the i_{th} risk factor.

4.4 Constructing the Implied, $IEP^1_{\tau,t}$ and $IEP^2_{\tau,t}$ eigenportfolios as trading strategies

Every factor, $\boldsymbol{F_i}$, can therefore be constructed as an eigenportfolio using $\boldsymbol{W_i}$ and forming N mutually independent portfolios, each represententing their own orthogonal dimension of risk. Together these portfolios fully explain the variation in the original asset space $\boldsymbol{R}$. Every element in $\boldsymbol{W_i}$ therefore describe the dollar amount to be invested in each of the original asset in order to form the ith eigenportfolio. Every risk dimension is therefore replicable as a real life investment strategy. These purely statistical risk factors form a basis for real life risk factors, as noted by (Lustig et al., 2011) among others. Their widely used common currency risk factors, the dollar factor (RX) and the carry factor (HML_{FX}), are highly correlated with their statistical counterparts. RX being an equally weighted level factor, explaining roughly 70% of the moves and the carry factor (HML_{FX}) being a interest rate sorted zero cost long/short slope factor, explaining roughly 15% of the remaining variance. The weighting of RX remain time invariant threw out time, but the HML_{FX} allocation is rebalanced monthly based on currencies forward premiums. Especially in the G10 currency space these risk factors are highly correlated (65% in our sample) with each other, forcing investors to strongly expose themselves on the general market sentiment, when investing on carry. Our aim is to construct the option implied RX and HML_{FX}, and especially focus on the amrket implied carry allocation. To accomplish the task, we proceed in the following way:

1. We back out implied correlation matrices in four different tenors (1m, 3m, 6m, 12m), utilizing the full cross-section of 45 different G10/USD ATM implied volatility crosses.

2. Implied correlation matrices are turned into implied covariance matrices and decomposed

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into eigenvectors (weight vectors).

3. Eigenvector matrix is regularized in a way that every negative element w_-^j in the eigenvector j is devided by the sum of all negative elements $\boldsymbol{abs}(\sum w_-^j)$ such that every eigenportfolio (=eigenvector) short leg sums up to -100% and every element in the long leg, w_+^j, sum up to 100%. Finally the long leg is devided by $\boldsymbol{max}(1, \boldsymbol{abs}(\sum w_-^j / \sum w_+^j))$ and the short leg by $\boldsymbol{max}(1, \boldsymbol{abs}(\sum w_+^j / \sum w_-^j))$, in order allow the original leverage in the allocation. This treatment does not effect the orthogonality condition, since the off diagonal elements of $\boldsymbol{W'W}$ still remain zero. It however does effect the diagonal, but that has no effect on our study. We also estimate the option implied zero-cost eigenportfolios, just by dropping out the last division part.

4. Since the sign of the eigenvector is arbitrary (since the required condition is only $\boldsymbol{W_j}'\boldsymbol{W_j} = 1$) , we can set $-\boldsymbol{W_j}$ without effecting the orthogonality conditions. Since both $\boldsymbol{W_i}$ and $-\boldsymbol{W_i}$ are valid eigenvectors, we have to ensure that the elements of first and second eigenportfolios take "logical" form. Since the first eigenportfolio is the market/-level factor, every currency on average should take positive weights. This is easy to ensure by forcing $max(\mathbf{1}\boldsymbol{W_1}, \mathbf{1}(-\boldsymbol{W_1})) > 0$. In order for the eigenportfolios to be orthogonal, some of the elements in the second eigenportfolio need to be negative. Since the second eigenportfolio resembles the carry strategyHML_{FX}, we force the second eigenvector weight's so, that the portfolio earns positive carry. In other words, the second eigenportfolio $IEP^2_{\tau,t}$ weights optimizing the carry (EPC_t) are determined by:

$$EPC_t = max(\boldsymbol{Car_t}\boldsymbol{W}_{\boldsymbol{IEP^2_{\tau,t}}}, \boldsymbol{Car_t}(-\boldsymbol{W}_{\boldsymbol{IEP^2_{\tau,t}}})) \tag{12}$$

where $\boldsymbol{Car_t} = \left[log\left(\frac{f_{t,i}}{s_{t,i}}\right)...log\left(\frac{f_{t,N}}{s_{t,N}}\right)\right]$ is $1xN$ vector of the 1 month carry for the N currencies and where $f_{t,i}$ is the 1 month forward and $s_{t,i}$ is the spot price for currency i. This way we have formed four sets(using $1m, 3m, 6m, 12m$ option data) of 1month forward looking weight vectors allowing long and short legs size to vary in $\boldsymbol{W}_{\boldsymbol{IEP^1_{\tau,t}}}$ and $\boldsymbol{W}_{\boldsymbol{IEP^2_{\tau,t}}}$. At the time of construction these portfolios are orthogonal, and we call them

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Implied RX $IEP^1_{\tau,t}$ and Implied HML_{FX}, $IEP^2_{\tau,t}$.

5. Finally, in order to examine further the nature of the new risk factors, we also create zero-cost carry factors $\boldsymbol{W}_{\boldsymbol{IEP}^{\boldsymbol{2,zc}}_{\boldsymbol{\tau,t}}}$ by eliminating the leverage between the short and the long leg and forcing them to become zero-cost. The allocation within the short and the long leg are therefore same as in $IEP^2_{\tau,t}$, but both long and short leg get the equal invested amount. These portfolios lack the the orthogonality property, although the dependencies remaining lower than between the traditional RX and HML_{FX}.

4.5 Variance riskpremiums for $\boldsymbol{IEP^1}$, $\boldsymbol{IEP^2}$ and for the common risk factors

We follow Della Corte et al. (2016) among others, to calculate variance risk premiums (VRP).

$$VRP_{t,\tau} = \mathbb{E}^{\mathbb{P}}(RV_{t,\tau}) - \mathbb{E}^{\mathbb{Q}}(RV_{t,\tau}) \tag{13}$$

Since the Bloomberg data is lacking the whole volatility surface information on all of the 45 currency crosses in G10 space, we construct implied volatilities for RX, HML_{FX}, IEP^1 and IEP^2 by replacing $\mathbb{E}^{\mathbb{Q}}(RV_{t,\tau})$'s model free variance, with the variance of delta neutral ATM straddles. The same approach is taken by Londono and Zhou (2017b) and Della Corte et al. (2016). The authors also compare the VRP results, estimating the risk-neutral expected variance with the model-free approach of Britten-Jones and Neuberger (2000), and conclude the ATM approach to be sufficient.

Implied variances of the portfolios are constructed using the traditional portfolio variance approach:

$$\mathbb{E}^{\mathbb{Q}}(RV_{t,\tau}) = (\boldsymbol{\omega_{t,\tau}} \boldsymbol{\Sigma_{t,\tau}} \boldsymbol{\omega_{t,\tau}}^T) \tag{14}$$

where $\boldsymbol{\omega_{t,\tau}}$ is a expected weight vector and $\boldsymbol{\Sigma_{t,\tau}}$ is the expected covariance matrix of the portfolio for the time period $\tau - t$. The implied variances of $IEP^1_{\tau,t}$ and $IEP^2_{\tau,t}$ can also be directly observed as $\boldsymbol{\lambda_i}$, where $i \in [1, 2]$..

Realized volatility, $\mathbb{E}^{\mathbb{P}}$, is proxied by the lagged realized volatility as in Bollerslev, Tauchen,

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and Zhou (2009) and Della Corte et al. (2016):

$$\mathbb{E}^{\mathbb{P}}(RV_{i,t,\tau}) = RV_{i,t-\tau,\tau} = \sqrt{\frac{252}{\tau}\sum_{k=0}^{\tau} r_{i,t-k}^2} \tag{15}$$

where r_t is a daily log return of implied eigenportfolio i, calculated as $\boldsymbol{RW'_{i,\tau,t}}$.

4.6 Term structure of Variance riskpremiums and the asset risk premium

A vast literature of VRP predicting power in different asset classes exists, but only a few studies concentrate on the VRP term structure. Della Corte et al. (2016) report the downward sloping shape of VRP term structure in FX markets, whereas Bormetti et al. (2016) demonstrate the relevance of the whole VRP term structure in explaining equity prices, and establish a theoretical relation between the equity risk premium, VRP and the slope of VRP term structure for one-component GARCH model. As the term structure contains information about the future dynamics of VRP over different time horizons, they show that the shape of the term structure (summarized by the slope) has strong predictive power on future stock-index returns. In the case where volatility follows GARCH(1,1) -process, they show the following link between the ERP, VRP and the ΔVRP:

$$ERP(t,\tau) = -\frac{c_1 + c_2 cov_t^{\mathbb{P}}(M_{t,\tau}, \epsilon_\tau^2) + c_3 VRP(t,\tau) + c_4 \Delta VRP}{2\eta\alpha\gamma} \tag{16}$$

where $ERP(t,\tau) = -cov_t^{\mathbb{P}}(M_{t,\tau}, \sqrt{h_\tau}\xi_\tau)$, $M_{t,\tau}$ is a stochastic discount factor, $c_1 = (\xi - 1)(\alpha + \omega)$, $c_2 = \xi\alpha$, $c_3 = 1 - (\beta + \alpha\gamma^2)$, $c_4 = 2$

4.7 Timing the HML_{FX} and IEP_τ^2 with VRP slope

As in Egbers and Swinkels (2015), we examine whether investors can utilize the predictability information of the $VRP_{t,6m}^1 - VRP_{t,1m}^1$ slope as an indicator to dynamically take either a long or short position in our 13 different currency carry trades portfolios (HML_{FX}, $IEP_\tau^{2,zc}$, IEP_τ^2 and $Orth_\tau$), compared to a static allocation. We take the following approach and condition our investment decision solely on 6m-1m VRP slope of the implied dollar eigenportfolio (IEP^1).

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We establish a long short setup, where we take along position in the 13 different carry trades when the timing indicator is above 0 threshold, and take a short position when it falls below 0 threshold:

$$rx_{i,t+1}^{dynamic} = signal_t * \boldsymbol{R_{t+1}} \boldsymbol{W'_{i,\tau,t}} \tag{17}$$

$$signal_t = \begin{cases} -1 & Threshold \leq 0 \\ 1 & Threshold > 0 \end{cases}$$

We denote the VTS timed strategies by HML_{FX}^{VTS}, $IEP_{t,\tau}^{2,VTS}$,$IEP_{t,\tau}^{2,ZC,VTS}$ and $Orth_{t,\tau}^{VTS}$.

5 Empirical results

5.1 The price of orthogonality vs zero-cost, characteristics of $IEP_{t,\tau}^2$

Table 11 presents the overall descriptive statistics for all 12 eigenportfolios. The first eigenportfolios repsesents the general market risk, implied RX factor, and there is little variation between the weights in different tenors threw out time, remaining close to a equally weighted portfolio. The second eigenportfolios, IEP_{τ}^2, represent the orthogonal version of implied HML_{FX}, whereas the zero-cost counterparts, $IEP_{t,\tau}^{2,ZC}$, are created by eliminationg the extra leverage. In the time of estimation the weights are normalized so that the elements of $IEP_{t,\tau}^1$ sum up to unity and $IEP_{t,\tau}^2$ elements so that the short and long legs sum up to unity within short and the long leg. The pearson correlation matrix between the implied eigenportfolios and the traditional risk factors can be seen in Table 10.

First we examine the decomposition of portfolio returns into carry and spot. The carry contributions are presented in Table 2. All $IEP_{t,\tau}^2$ and $IEP_{t,\tau}^{2,ZC}$ eigenportfolios earn positive carry by construction. The traditional HML_{FX} with 24bp's per month is the best carry earning portfolio in the group, where as the orthogonal $IEP_{t,\tau}^2$ portfolios contribute from 13bp to 15bp per month and the zero-cost $IEP_{t,\tau}^{2,ZC}$'s 17bp to 20bp. The " carry cost of orthogonality", i.e. $IEP_{t,\tau,carry}^{2,zc} - IEP_{t,\tau,carry}^{2}$ is therefore -4bp in all tenors. So in other words, moving from zero-cost portfolio (correlated with $IEP_{\tau,t}^1$) to orthogonal (wrt $IEP_{\tau,t}^1$)

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reduces slope of the carry drift by 4bp/month.

Next, the spot price contributions are presented in Table 1. Its worth noticing, that the worst performer in the group wrt negative spot average return and lowest Sharpe ratio is the HML_{FX}. There exist a large body of literature proving the profitability of HML_{FX} to exist solely due to the positive yield differential returns, whereas spot price is shown to have no contributions in the long run. The spot contribution results for the best performing $IEP^{2,ZC}_{t,\tau}$'s are very surprising. By contrast, the option implied carry strategy allocation seems to be able to not just obtain equally high carry as HML_{FX}, but it is also able to select the currencies with high appreciation potential. The Sharpe ratios for the monthly spot returns are between 0.2 to 0.37 (-0.09 for HML_{FX}) and the average monthly spot returns between 15bp to 28bp per month (-7bp for HML_{FX}), the 6m and 12m strategies being the star performers. The spot price distributions of all $IEP^{2,ZC}_{t,\tau}$ experience modest negative skewness ranging from -0.49 to -1.44, where as the respective HML_{FX} figure is -0.96. The spot price contribution of the orthogonal $IEP^{2}_{t,\tau}$ are more modest. The 1m and 3m portfolios actually contributes negatively on the overall returns (-7bp and -2bp per month respectively), where as 6m and 12m have a positive spot contribution of 5bp and 12bp per month. All $IEP^{2}_{t,\tau}$ spot return distributions, except 1m, experience small negative skewness.

Table 1. This table presents the spot price descriptive statistics of all eight implied carry eigenportfolios, (IEP^2_τ). Portfolios are rebalanced monthly and their allocations are based on weight vectors calculated using the spectral decomposition of option markets forward looking implied correlation matrices with 1m, 3m, 6m and 12m maturities. Portfolios are established based on these weights at t, and their performance is evaluated at $t+1$. All the figures are monthly log-spotprice changes of he respective portfolios, spanning from 09/2008 to 08/2016.

	$HML_{fx,spot}$	$IEP^2_{spot,1m}$	$IEP^2_{spot,3m}$	$IEP^2_{spot,6m}$	$IEP^2_{spot,12m}$	$IEP^{2,zc}_{spot,1m}$	$IEP^{2,zc}_{spot,3m}$	$IEP^{2,zc}_{spot,6m}$	$IEP^{2,zc}_{spot,12m}$
Mean	-0.0007	-0.0007	-0.0002	0.0005	0.0012	0.0020	0.0015	0.0022	0.0028
Std	0.0272	0.0226	0.0204	0.0203	0.0206	0.0299	0.0260	0.0257	0.0260
Skewness	-0.96	0.30	-0.06	-0.36	-0.65	-0.49	-1.02	-1.15	-1.44
Kurtosis	4.25	1.49	1.00	2.10	3.90	7.08	5.20	5.66	7.93
Sharpe	-0.09	-0.11	-0.03	0.09	0.20	0.23	0.20	0.30	0.37

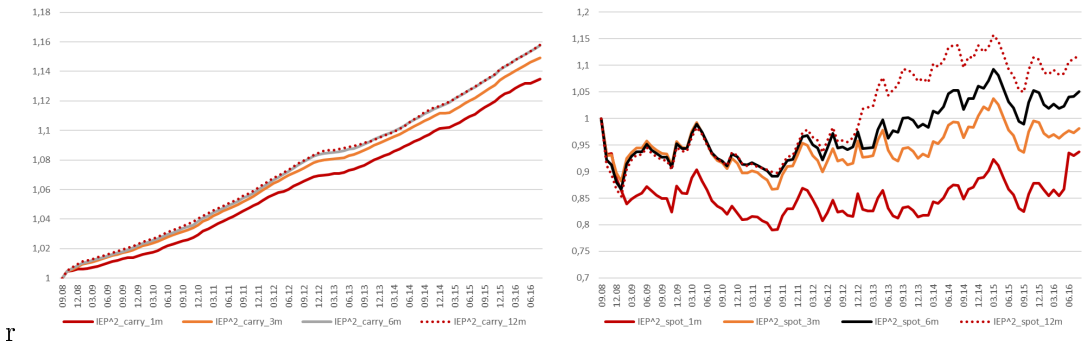
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Table 2. This table presents the descriptive statistics of 1 month carry on all eight implied carry eigenportfolios, (IEP^2_τ). Portfolios are rebalanced monthly and their allocations are based on weight vectors calculated using the spectral decomposition of option markets forward looking implied correlation matrices with 1m, 3m, 6m and 12m maturities. Portfolios are established based on these weights at t, and their performance is evaluated at $t+1$. All the figures are monthly $log(f_t/s_t)$ changes of he respective portfolios, spanning from 09/2008 to 08/2016.

	$IEP^2_{carry,1m}$	$IEP^2_{carry,3m}$	$IEP^2_{carry,6m}$	$IEP^2_{carry,12m}$	$IEP^{2,zc}_{carry,1m}$	$IEP^{2,zc}_{carry,3m}$	$IEP^{2,zc}_{carry,6m}$	$IEP^{2,zc}_{carry,12m}$	HML_{carry}
Mean	0.0013	0.0015	0.0015	0.0015	0.0017	0.0019	0.0020	0.0020	0.0024
Std	0.0007	0.0006	0.0006	0.0006	0.0008	0.0007	0.0007	0.0007	0.0004
Skewness	1.16	1.20	1.33	1.39	1.03	0.72	0.77	0.62	0.98
Kurtosis	6.77	7.40	8.24	8.62	8.57	7.02	7.02	6.64	1.34

Table 3. This table presents the total return (spot + carry) descriptive statistics of all eight implied carry eigenportfolios, (IEP^2_τ). Portfolios are rebalanced monthly and their allocations are based on weight vectors calculated using the spectral decomposition of option markets forward looking implied correlation matrices with 1m, 3m, 6m and 12m maturities. Portfolios are established based on these weights at t, and their performance is evaluated at $t+1$. All the figures are monthly log-spotprice changes of he respective portfolios, spanning from 09/2008 to 08/2016.

	HML_{fx}	IEP^2_{1m}	IEP^2_{3m}	IEP^2_{6m}	IEP^2_{12m}	$IEP^{2,zc}_{1m}$	$IEP^{2,zc}_{3m}$	$IEP^{2,zc}_{6m}$	$IEP^{2,zc}_{12m}$
Mean	0.0011	0.0001	0.0013	0.0021	0.0027	0.0037	0.0034	0.0042	0.0048
Std	0.0277	0.0244	0.0203	0.0201	0.0203	0.0296	0.0258	0.0254	0.0257
Skewness	-0.90	-0.59	-0.01	-0.30	-0.56	-0.44	-0.95	-1.09	-1.36
Kurtosis	3.39	4.80	0.80	1.81	3.48	6.65	4.77	5.26	7.43
Sharpe	0.13	0.02	0.22	0.35	0.46	0.44	0.45	0.57	0.64



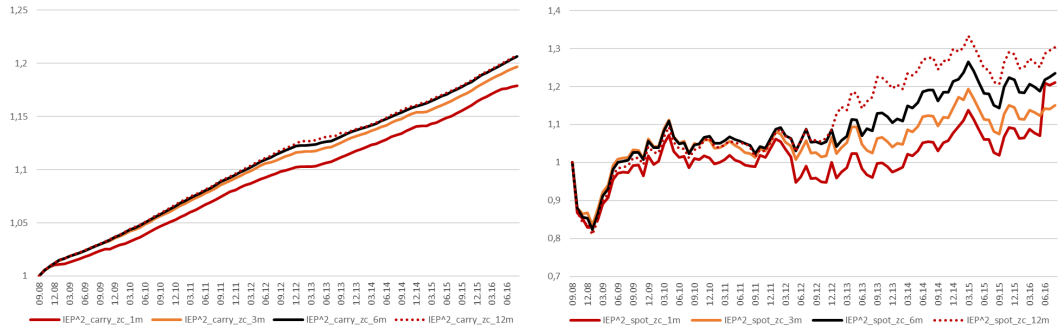
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(a) The carry contribution of IEP^2_τ

(b) The spot contribution of IEP^2_τ

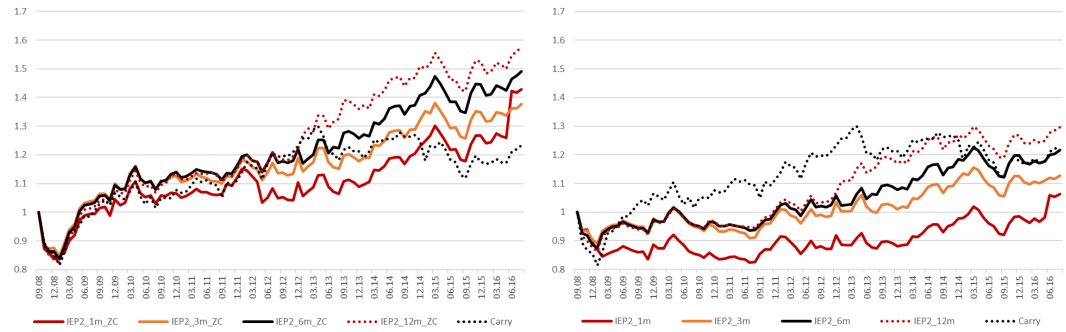
Figure 1. The carry and spot contributions for IEP^2_τ

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(a) The carry contribution of $IEP_\tau^{2,zc}$ (b) The spot contribution of $IEP_\tau^{2,zc}$

Figure 2. The carry and spot contributions for $IEP_\tau^{2,zc}$



(a) Total return on $IEP_\tau^{2,zc}$ (b) Total return on IEP_τ^2

Figure 3. The total returns for $IEP_\tau^{2,zc}$ and IEP_τ^2

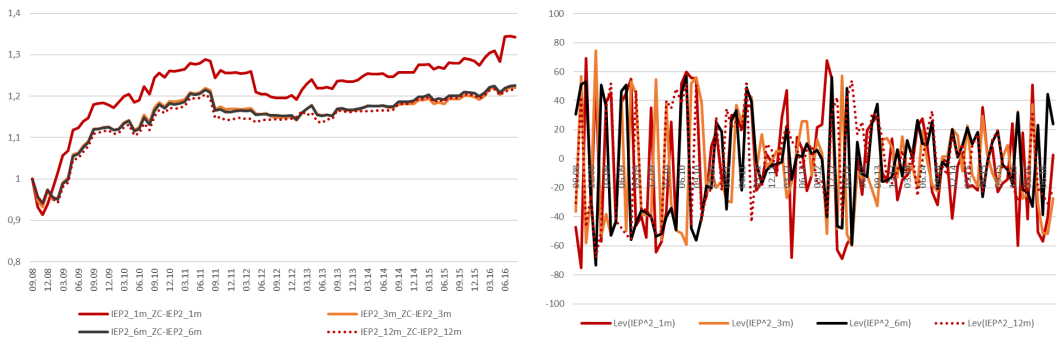
Finally, we concentrate on the difference between the zero-cost and orthogonal IEP^2's, $(Orth_\tau = IEP_\tau^{2,zc} - IEP_\tau^2)$. This measure is short on the extra leverage of the IEP^2factor, generated by the PCA. This leverage is needed to convert the zero-cost carry factor, $IEP_\tau^{2,ZC}$ to orthogonal IEP_τ^2 with respect to IEP_τ^1. The time variation of this extra leverage is illustrated by Figure 4 (b). This leverage fluctuates between -0.80 $<Lev_\tau <$0.80 in all times and averages at -3.2 %, -5.1 %, -2.5 % and -1.7% with τ 1m, 3m, 6m and 12m, respectively. $IEP_{\tau,t}^2$ are therefore slightly net short on avarage, where as the boundaries for upper and lower quintile are -0.32 $<Lev_\tau <$0.26 in all tenors. So on average only modest extra leverage compared to zero-cost portfolios is needed to maintain the orthogonality between the first and second eigenvectors in same tenor. We illustrate return profile of $Orth_{t,\tau}$ in Tabel 4, where we control the results with the traditional RX and HML_{FX} risk factors. The regression results indicate

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a heavy loading on RX factor, but the strategy produces highly statistically significant alpha's between 20bp to 50bp per month. Also it is intresting to note that all of the $Orth_{t,\tau}$ experience positive skewness (between 0.57 to 0.1) compared to -0.9 of HML_{FX}. Also the sharpe ratios are dramatically improved compared to the HML_{FX} -factor, since the Sharpe ratio of $Orth_{t,\tau}$ differes between 0.87 and 0.53, where as corresponding figure for HML_{FX} is 0.13.

Table 4. Zero-cost vs orthogonal IEP_τ^2 portfolio generates consistent alpha(20bp-50bp/month). Strategy is highly exposed to RX but not to HML_{FX}

	$\boldsymbol{Orth_{1m}}$	$\boldsymbol{Orth_{3m}}$	$\boldsymbol{Orth_{6m}}$	$\boldsymbol{Orth_{12m}}$
$\boldsymbol{RX}$	0.236	0.324	0.312	0.310
	(4.134)***	(7.695)***	(7.781)***	(7.885)***
$\boldsymbol{HML_{FX}}$	0.075	0.081	0.066	0.067
	(1.289)	(1.876)*	(1.606)	(1.654)
$\boldsymbol{Alpha}$	0.0040	0.0020	0.0020	0.0050
	(2.923)***	(2.357)**	(2.54)***	(2.484)***
$\boldsymbol{R^2}$	0.29	0.58	0.57	0.58



(a) Returns for the $Orth_\tau = IEP_\tau^{2,zc} - IEP_\tau^2$

(b) IEP_τ^2 leverage (%)

Figure 4. $Orth_\tau$

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Table 5. This table presents the descriptive statistics of 1 month returns on selected eigenportfolios, ($IEP_{\tau}^{2,ZC}$) and $Orth_{\tau}$. Portfolios are rebalanced monthly and their allocations are based on weight vectors calculated using the spectral decomposition of option markets forward looking implied correlation matrices with 1m, 3m, 6m and 12m maturities. Portfolios are established based on these weights at t, and their performance is evaluated at $t+1$. All the figures are monthly $log(f_t/s_t)$ changes of he respective portfolios, spanning from 09/2008 to 08/2016.

	$Orth_{1m}$	$Orth_{3m}$	$Orth_{6m}$	$Orth_{12m}$	HML_{fx}	$IEP_{1m}^{2,zc}$	$IEP_{3m}^{2,zc}$	$IEP_{6m}^{2,zc}$	$IEP_{12m}^{2,zc}$
Mean	0.0036	0.0021	0.0021	0.0020	0.0011	0.0037	0.0034	0.0042	0.0048
Std	0.0143	0.0137	0.0129	0.0128	0.0277	0.0296	0.0258	0.0254	0.0257
Skewness	0.57	0.10	0.35	0.20	-0.90	-0.44	-0.95	-1.09	-1.36
Kurtosis	2.31	5.56	4.97	5.18	3.39	6.65	4.77	5.26	7.43
Sharpe	0.87	0.53	0.57	0.55	0.13	0.44	0.45	0.57	0.64

5.2 Predictive power of implied eigenportfolio variance risk premiums

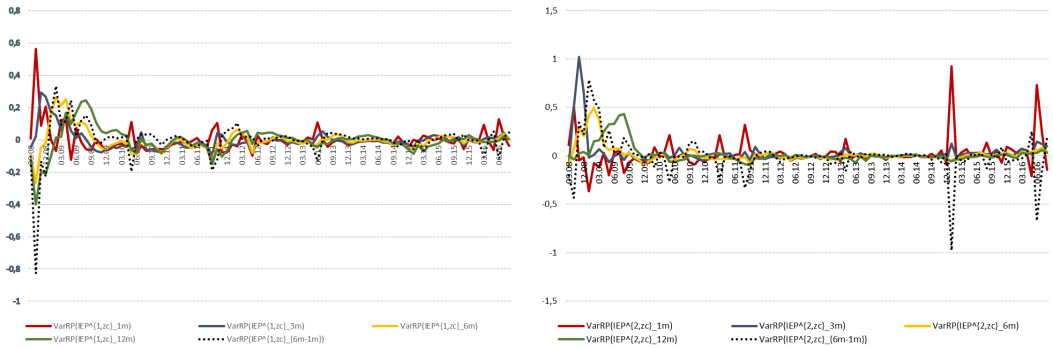
We follow Londono and Zhou (2017b) to establish our benchmark empirical regression setup for the predictive power of currency variance risk premiums. Our empirical analysis begins with ordinary least squares (OLS) estimation of predictive regressions for a one-month ahead horizon. The first predictive regression provides a simple way for assessing the intertemporal risk-return trade off in FX risk factors. Our empirical regression setup is as follows

$$rx_{i,t+1} = \beta_1 + \beta_2(VRP_{t,6m}^{1} - VRP_{t,1m}^{1}) \tag{18}$$

where $rx_{i,t+1}$ is the excess return on 12 implied eigenportfolio retuns, and the slope of variance risk premium for the first implied eigenportfolio (IEP^1), and $VRP_{t,Tau}^{1}$'s are the variance risk premiums for the first option implied eigenportfolio at $t-1$. Table 12 reports the predictive power regression results for the variance risk premium slope measures over 1 month ahead. Our results suggest that the first eigenportfolio VRP term structure plays a crucial role in explaining the future appreciation rate for both first and second zero-cost and orthogonal IEP's in all maturities. All but two of the sixteen regressions have statistically significant loading on the VRP slope. Descriptive statistics for the VRP's are reported in Table 13. Londono and Zhou (2017b) report individual USD/X currency VRP's to experience high volatility, negative skewness and high kurtosis. Since our currency implied eigenportfolios are portfolios with currencies quoted as G10/USD, we experience positive skewness in $1m, 3m$

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and $6m$ IEP^1 VRP's, but slight negative skewness on $12m$ tenor. All four tenors report high kurtosis. The results for $IEP^{2,zc}_{t,\tau}$ are particularly interesting. The β's of the VRP slope has statistical significance of $<1\%$ in 3m to 12m and $<5\%$ in 1m tenor. The magnitude ranges from 0.70 to 0.96 and seems to be linearly increasing. The constant is found to be statistically significant only in 6m and 12m tenors.



(a) Annualized 1m, 3m, 6m and 12m VRP's for IEP^1_τ

(b) Annualized 1m, 3m, 6m and 12m VRP's for $IEP^{2,zc}_\tau$

Figure 5. Annualized 1m, 3m, 6m and 12m VRP's for $IEP^1_{t,\tau}$ and $IEP^{2,zc}_{t,\tau}$

Table 6. The predictive power of the implied RX 's 6m-1m variance risk premium slope for the implied eigenportfolio returns. This table reports the estimated coeffcients for the following regressions: $rx_{i,t+1} = \beta_1 + \beta_2(VRP^1_{t,6m} - VRP^1_{t,1m})$,where $rx_{i,t+1}$ is the excess return on 16 implied eigenportfolio retuns, and the slope of variance risk premium for the implied RX, (IEP^1), the difference between 6 month and 1 month VRP's at $t-1$. The standard errors are corrected by Newey-West with 12 lags and the standard deviations are reported in parenthesis. and *** represent the usual 10, 5 and 1% significance levels. The sample period runs from Sep 2008 to Aug 2016. Slope VRP in the table stands for $VRP(IEP^1_{6m}) - VRP(IEP^1_{1m})$ and $Ortho_\tau$ stands for the difference $IEP^{2,zc}_\tau - IEP^2_\tau$) between zero-cost and orthogonal(leveraged) implied carry eigenportfolio.

	IEP^1_{1m}	IEP^2_{1m}	IEP^1_{3m}	IEP^2_{3m}	IEP^1_{6m}	IEP^2_{6m}	IEP^1_{12m}	IEP^2_{12m}
Slope VRP	0.709	0.325	0.708	0.340	0.667	0.402	0.649	0.516
	(2.18)**	(1.230)	(2.17)**	(1.561)	(2.04)**	(1.87)*	(1.973)*	(2.40)**
constant	-0.0003	0.0002	-0.0002	0.0014	-0.0002	0.0022	-0.0001	0.0020
	(-0.008)	(0.009)	(-0.06)	(0.661)	(-0.049)	(1.06)	(-0.004)	(1.41)
R^2	0.04	0.01	0.04	0.02	0.03	0.26	0.03	0.05

	$IEP^{2,ZC}_{1m}$	$IEP^{2,ZC}_{3m}$	$IEP^{2,ZC}_{6m}$	$IEP^{2,ZC}_{12m}$	$Orth_{1m}$	$Orth_{3m}$	$Orth_{6m}$	$Orth_{12m}$
Slope VRP	0.703	0.806	0.860	0.955	0.378	0.465	0.457	0.439
	(2.23)**	(2.996)***	(3.273)***	(3.638)***	(2.501)**	(3.302)***	(3.443)***	(3.334)***
constant	0.0040	0.0036	0.0044	0.0050	0.0037	0.0022	0.0023	0.0022
	(1.325)	(1,41)	(1.786)*	(2.030)**	(2.602)**	(1.671)*	(1.804)*	(1.742)*
R^2	0.04	0.08	0.09	0.12	0.05	0.10	0.10	0.10

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5.3 Statistical methodology and the results for Sharpe ratio comparison

We follow the notation in Ledoit and Wolf (2008) and Bekaert and Panayotov (2015), and denote the sample average returns of a HML_{FX} and one of the implied eigenvector strategies IEP^2_τ as μ_{HML} and $\mu_{IEP^2_\tau}$ and their respective uncentered sample second moments as σ^2_{HML} and $\sigma^2_{IEP^2_\tau}$. Also let $\hat{v} = [\mu_{HML}, \mu_{IEP^2_\tau}, \sigma^2_{HML}, \sigma^2_{IEP^2_\tau}]$ and we assume that $\sqrt{T}(v - \hat{v}) \xrightarrow{d} (0, \Omega)$, where v is the population counterpart of $\hat{v}$, T is sample length and Ω is HAC Newey-West covariance matrix Newey and West (1987) with Quadratic-Spectral kernell with bandwidth m. The population deviations of the strategies Sharpe ratios from their sample difference take the following form:

$$\hat{\Delta} = f(\hat{v}) = \frac{\mu_{HML}}{\sqrt{\sigma^2_{HML} - \mu^2_{HML}}} - \frac{\mu_{IEP^2_\tau}}{\sqrt{\sigma^2_{IEP^2_\tau} - \mu^2_{IEP^2_\tau}}} \tag{19}$$

$$\sqrt{T}(\Delta - \hat{\Delta}) \xrightarrow{d} (0, \nabla' f(\hat{v}) \Omega \nabla f(\hat{v}) \tag{20}$$

Where

$$\nabla' f(a, b, c, d) = \Big(\frac{c}{(c - a^2)^{1.5}}, -\frac{d}{(d - b^2)^{1.5}}, -\frac{a}{2(c - a^2)^{1.5}}, \frac{b}{2(d - b^2)^{1.5}} \Big) \tag{21}$$

And since $\hat{\Omega}$ is a consistent estimator of Ω, we can express the standard error of the $\hat{\Delta}$:

$$s(\hat{\Delta}) = \frac{\nabla' f(\hat{v}) \hat{\Omega} \nabla f(\hat{v}}{T} \tag{22}$$

For the null hypothesis testing, we follow Ledoit and Wolf (2008) and bootstrap the Implied eigenportfolio strategies against the benchmark HML_{FX}, and consider the studentized random variable, $L = \frac{\hat{\Delta} - \Delta^*}{s(\Delta^*)} s(\hat{\Delta}) \leq 0$ where Δ^* is a Sharpe ratio difference between the strategies, computed with bootstrapped returns with the corresponding standard error $s(\Delta^*)$. We examine the lack of autocorrelation in our sample, but utilize the algorithm of Ledoit and Wolf (2008) to find the optimal block size in our bootstrapping algorithm. This optimized block size is reported on the Table 8 I.e. we utilize the pairwise bootstrap method with 5000 samples with optimal block size and with replacement.

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And finally the one sided p-value for the null $H_0 : \Delta = 0$ is:

$$\hat{p} = \Phi\Big(- \frac{|\hat{\Delta}|}{s(\hat{\Delta})} \Big) \tag{23}$$

Where $\Phi()$ is the cdf of the standard normal distribution.

Sharpe ratio results are reported in Table 7, where two kind of p-values are reported, the traditional HAC and pre-whitened HAC. Statistically significant Sharpe ratios are found in $IEP_{6m}^{2,ZC}$, $IEP_{12m}^{2,ZC}$, $IEP_{12m}^{2,ZC,VTS}$, HML_{FX}^{VTS} and $Orth_{12m}^{VTS}$. We confirm the found results by utilizing the above determined bootstrapping method. Those results can be found in Table 8.

Table 7. This table presents the p-values for the HML_{FX} and option implied carry eigenportfolio monthly Sharpe ratio differences, $\hat{\Delta}$. These alternative carry strategies are based on a market allocations of a second option implied eigenvector IEP_{τ}^{2}, $IEP_{\tau}^{2,ZC}$, $Orth_{\tau}$ and their VTS timed counterparties in four different tenors (1m, 3m, 6m and 12m). $\hat{\Delta}$ equals the Sharpe ratio difference between $HML_{FX} - Alternative$.

	IEP_{1m}^{2}	IEP_{3m}^{2}	IEP_{6m}^{2}	IEP_{12m}^{2}	$IEP_{1m}^{2,ZC}$	$IEP_{3m}^{2,ZC}$	$IEP_{6m}^{2,ZC}$	$IEP_{12m}^{2,ZC}$	$Orth_{1m}$	$Orth_{3m}$	$Orth_{6m}$
$\hat{\Delta}$	0.04	0.00	-0.04	-0.07	-0.06	-0.06	-0.10	-0.12	-0.13	-0.09	-0.10
$p - val_{HAC}$	0.72	0.97	0.66	0.40	0.40	0.30	0.13	0.06*	0.19	0.32	0.27
$p - val_{PW}$	0.71	0.97	0.65	0.36	0.37	0.26	0.09*	0.03**	0.22	0.32	0.28
SE	0.10	0.08	0.08	0.07	0.07	0.06	0.06	0.05	0.10	0.09	0.09
SE_{PW}	0.04	0.00	-0.04	-0.07	-0.06	-0.06	-0.10	-0.12	-0.13	-0.09	-0.10

	$Orth_{12m}$	$IEP_{1m}^{2,ZC,VTS}$	$IEP_{3m}^{2,ZC,VTS}$	$IEP_{6m}^{2,ZC,VTS}$	$IEP_{12m}^{2,ZC,VTS}$	HML_{FX}^{VTS}	$Orth_{1m}^{VTS}$	$Orth_{3m}^{VTS}$	$Orth_{6m}^{VTS}$	$Orth_{12m}^{VTS}$
$\hat{\Delta}$	-0.09	-0.08	-0.11	-0.11	-0.17	-0.19	-0.09	-0.13	-0.15	-0.19
$p - val_{HAC}$	0.32	0.61	0.47	0.47	0.22	0.18	0.52	0.36	0.27	0.14
$p - val_{PW}$	0.32	0.49	0.28	0.28	0.09*	0.07*	0.39	0.21	0.14	0.04**
SE	0.09	0.11	0.10	0.10	0.10	0.10	0.11	0.10	0.10	0.09
SE_{PW}	-0.09	-0.08	-0.11	-0.11	-0.17	-0.19	-0.09	-0.13	-0.15	-0.19

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Table 8. This table reproduces Table 7, with bootstrap-based p-values, $p-val_{HAC}$, with prewhitended HAC standard error for differences in Sharpe ratios $\hat{\Delta}$. Test compares the HML_{FX} for the option implied carry portfolio monthly Sharpe ratio differences. These alternative carry strategies are based on a market allocations of a second option implied eigenvector IEP^2_τ, $IEP^{2,ZC}_\tau$, $Orth_\tau$ and their VTS timed counterparties. $\hat{\Delta}$ equals the Sharpe ratio difference between HML_{FX}. Our data includes 96 monthly observations and spans from 09/2008 to 08/2016. Column $p-val_{HAC}$ in tha table denoted the p-values from the test based on pairwisely randomized returns, using again the studentized random variable $L = \frac{\hat{\Delta}-\Delta^*}{s(\Delta^*)} s(\hat{\Delta}) \leq 0$ as in denoted in chapter "Statistical methodology for Sharpe ratio comparison". $t-val$ denotes the absolute normalized value of the Sharpe Ratio differences in the original data and finally *blocksize* denotes the number of returns drawn once. In other words, the lack of autocorrelation in the return seriesses allows us to use the i.i.d. assumption and execute pair-wise drawings with replacements of one observation at a time.

	IEP^2_{1m}	IEP^2_{3m}	IEP^2_{6m}	IEP^2_{12m}	$IEP^{2,ZC}_{1m}$	$IEP^{2,ZC}_{3m}$	$IEP^{2,ZC}_{6m}$	$IEP^{2,ZC}_{12m}$	$Orth_{1m}$	$Orth_{3m}$	$Orth_{6m}$
$p-val_{HAC}$	0.36	0.48	0.33	0.18	0.19	0.14	0.05**	0.02**	0.11	0.17	0.14
$\hat{\Delta}$	0.04	0.00	-0.04	-0.07	-0.06	-0.06	-0.10	-0.12	-0.13	-0.09	-0.10
t-val	0.37	0.04	0.45	0.92	0.89	1.13	1.69	2.19	1.23	0.99	1.08
block size	1	1	1	1	1	1	1	1	1	1	1

	$Orth_{12m}$	$IEP^{2,ZC,VTS}_{1m}$	$IEP^{2,ZC,VTS}_{3m}$	$IEP^{2,ZC,VTS}_{6m}$	$IEP^{2,ZC,VTS}_{12m}$	HML^{VTS}_{FX}	$Orth^{VTS}_{1m}$	$Orth^{VTS}_{3m}$	$Orth^{VTS}_{6m}$	$Orth^{VTS}_{12m}$
$p-val_{HAC}$	0.16	0.25	0.15	0.16	0.05**	0.04**	0.20	0.12	0.08*	0.03**
$\hat{\Delta}$	-0.09	-0.08	-0.11	-0.11	-0.17	-0.19	-0.09	-0.13	-0.15	-0.19
t-val	1.00	0.68	1.08	1.07	1.69	1.83	0.86	1.25	1.48	2.05
block size	1	1	1	1	1	1	1	1	1	1

5.4 Stochastic dominance (SD) results to compare the ultimate preferences between strategies

In the presence of skewed return distributions, the Sharpe-Ratio test itself might turn out to be inadequate. The stochastic dominance framework offers an utility-based method for evaluating choices under uncertainty, which are not tight to distribution symmetry assumptions. It can quantify a rational choise between two risky prospects, whose distributions may be highly non-normal. If strategy A 's returns always exceed that of B, no rational investor will choose B since he is always better off with A. A is said to stochastically dominate B at first order. First order stochastic dominance (FSD), however, is a very strong result, as it implies that all rational investors will prefer A over B, regardless of their risk preferences. Since the FSD criterion may be too restricting, the empirical work has concentrated on second order stochastic dominance (SSD) and third order stochastic dominance (TSD). These more general concepts apply to choices made by non-satiated and risk-averse individuals. Since SSD nor TSD make assumptions about the exact shape of an individual's utility function, any rational investor with with concave utility function would choose A over B, in the case of SSD-dominance. Since it is natural to argue the rational investors to prefer a strategy with more positively skewed

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return distribution, TSD includes a preference for positive skewness to non-satiation and risk aversion assumptions. So a TSD individual would choose a positively skewed strategy among two investments with same mean variance. TSD dominance can however also be caused by higher mean and/or variance aswell.

We follow the stochastic dominance test setup of Linton et al. (2005) and define that the SSD and TSD between the benchmark HML_{FX} and an alternative strategy, can be based on comparing functions of the cumulative return distributions $F_k(x)$ of the different investment strategies k. A practical characterization of any SD order is due to Davidson and Duclos (2000). Strategy X_1 stochastically dominates strategy X_2 at order s, if $D_1^{(s)}(x) \leq D_2^{(s)}(x)$ for all x and with strict inequality for some x, where,

$$D_k^{(s)}(x) = \int_{-\infty}^{x} D_k^{(s-1)}(x) \tag{24}$$

$$D_k^{(1)}(x) = F_k(x) \tag{25}$$

for all x and with strict inequality for some x,

The LMW test evaluetaes the following two hypotheses:

$H_0 : D_k^s(x) \leq D_z^s(x)$ for all $x \in V$

and

$H_1 : D_k^s(x) > D_z^s(x)$ for some $x \in V$

where $\{Z_i\}, i = 1, 2, ..., N$ is a random sample from a population with CDF $F_Z(x)$ and $D_z^s(x)$ is determined analogously to Equation (24) and Equation (25), and finally V is the joint support for the two distributions. If H_0 is that k dominates z at the sth order, then the H_1 is that SD fails at some point. The null hypothesis has to be tested in two steps. First $H_0^1 : x \succ_s z$, where $\succ_s$ indicates s^{th} order dominance. The second null is $H_0^2 : z \succ_s x$. We conclude that x dominates z if we accept H_0^1 and reject H_0^2. Finally, to test the previously described two null hypotheses, we use the following test statistic proposed by Linton et al. (2005):

$$\overline{LMW^s} = \sup_x \sqrt{N}[\hat{D}_z^s(x) - \hat{D}_k^s(x)] \tag{26}$$

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where N is the sample size. Linton et al. (2005) show that the asymptotic null distribution of the test statistic is non-standard and propose bootstrapping with subsampling (to capture the non i.i.d. features) to compute the empirical p-values of the test.

$$\overline{LMW_i^s} = \sup_x \sqrt{b}[\hat{D}_{z,1}^s(x) - \hat{D}_{k,1}^s(x)] \tag{27}$$

for $i = 1, 2, ..., N - b + 1$

The test statistic takes the following form:

$$\hat{p_s} = \frac{1}{N-b+1} \sum_{i=1}^{N-b+1} 1\big(\overline{LMW_i^s} - \overline{LMW^s} > 0\big) < \alpha \tag{28}$$

Asymptotic theory requires that the sub-sample size b should increase with N, with the ratio b/N tending to zero as N tends towards infinity, but does not provide more specific guidance. As in Fong (2010), we choose $b \in [1, 40]$. As in LMW, the statistical inference is based on the mean value of the 40 bootsrapped p-values.

Since our aim is to test wether option implied information can produce improved alternatives for the benchmark HML_{FX}-factor returns, our focus is on the p-values are based on one-sided bootstrap confidence interval for Sharpe Ratio and stochastic dominance test that also count the skewness differences.

We run our our two-stage H_0 hypothesis test against the benchmark HML_{FX} factor returns, which has been performing relatively poorly in post-financial crisis period of low interes rate differentials. The results for two-stage FSD, SSD and TSD tests are presented in Table 9, where as Figures 7 and 8 graphically illustrate the first, second and third order dominances for selected strategies. The results for the FSD test are reported in first two columns of the Table 9. The first column indicates $H_0^1 : HML_{FX} \succ_s z$ to be rejected for $IEP_{6m}^{2,ZC}$ and $IEP_{12m}^{2,ZC}$ zero-cost portfolios, for all four $Orth_\tau$ and for all four VRP term structure timed (VTS) $Orth_\tau^{VTS}$. The second null is $H_0^2 : z \succ_2 HML_{FX}$ is accepted only for $IEP_{6m}^{2,ZC}$, $IEP_{12m}^{2,ZC}$ and $Orth_{3m}^{VTS}$, proving these three portfolios to FSD dominate HML_{FX} at 5% significance level. This is a really strong result, stating that the second zero-cost eigenportfolio of the 6m

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and 12m option implied covariance matrix contains crucial information on the carry portfolio allocation for a 1m holding period. Regardless of their risk preferences, all rational investors will prefer $IEP_{6m}^{2,ZC}$ and $IEP_{12m}^{2,ZC}$ over HML_{FX}.

Results for the SSD tests are in shown in the third and fourth panels of Table 9. The null hypothesis $H_0^1 : HML_{FX} \succ_2 z$, is now widely rejected. All four $Orth_\tau$ and their VTS timed counterparties, $Orth_\tau^{VTS}$ all reject the $H_0^1 : HML_{FX} \succ_2 z$ at 1% level, where as VTS timed zero-cost second eigenportfolios, $IEP_\tau^{2,ZC,VTS}$, all reject the first null at either at 10% or 5% level. Also the VTS timing seems to have a significant predicting power on HML_{FX} itself, since HML_{FX}^{VTS} rejects the first null at 5% level. Since the second null, $H_0^2 : z \succ_2 HML_{FX}$, in column four is accepted in all cases, all $Orth_\tau$, $Orth_\tau^{VTS}$, $IEP_\tau^{2,ZC,VTS}$ and HML_{FX}^{VTS} end up SSD dominating the traditional HML_{FX} strategy.

Finally the TSD test in Panels 5 and 6 in Table 9 do not add any more dominating strategies to our list, but they confirm the same results as the SSD table.

Overall, the FSD-test results indicate that all rational investors will prefer $IEP_{6m}^{2,ZC}$, $IEP_{12m}^{2,ZC}$ and $Orth_{3m}^{VTS}$ over HML_{FX}, regardless of their risk preferences, whereas all non-satiated and risk-averse individuals with concave utility function would choose $Orth_\tau$, $Orth_\tau^{VTS}$, $IEP_\tau^{2,ZC,VTS}$ and HML_{FX}^{VTS} over HML_{FX}, indicated by SSD- and TSD-dominance. These results confirm the natural asumption, that the rational investors prefer a strategy with more positively skewed return distribution. The high returns of implied eigenportfolios are also unlikely to be explained by risk. Finally, the strong rejection of the second null hypothesis for the portfolios supports the value of diversification in carry trades. This result is consistent with that of Darvas (2009) for currency carry trades.

First, the option markets seems to contain some crucial information regarding the risk factor portfolio allocation in G10 currencies. Even though the IEP_τ^2 profitability seems to be inferior with respect to their traditional counterparty, HML_{FX}, the stochastic dominance tests are unable to indicate any statistically significant dominance structure at any (SD) order between any of the four implied eigenportfolio and HML_{FX}. Benefits of these option implied risk factors seem to exist in the diversification front. As the RX and HML_{FX} risk

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factors experience stunning 60% correlation over the whole observation period among the G10 currencies, the respective realized correlation between IEP_τ^2 and IEP_τ^1 varies only between 10% and 21%. The large RX exposure seems to be driving the carry profits of HML_{FX}, and deteriorate the diversification effect. Constructed to be orthogonal with respect to IEP_τ^1, option implied carry factors seem to maintain their independence fairly well.

Second, we analyze the option implied zero-cost carry factors $IEP_\tau^{2,ZC}$ and reveal very interesting results. These eigenportfolios contain the same allocation within the long and short legs as the IEP_τ^2, but the extra leverage (needed for the orthogonality) is removed and the eigenportfolios are forced to become zero-cost. These portfolios experience 69% to 81% correlation with the HML_{FX} over the observed period. The FSD tests report very strong results for the superiority of $IEP_{6m}^{2,ZC}$and $IEP_{12m}^{2,ZC}$ over the carry factor, where as in the 3m and 1m tenors, neither FSD, SSD nor TSD are able to detect any dominance order between them. More interestingly, the traditional carry trade profitability is widely known to be driven by the positive interest rate differential, while the spot price tend not to have no contribution on the returns in the long run. Table 2 reports the carry contributions of the different portfolios. The rate differential contributed HML_{FX} monthly 24bp, where as $IEP_{6m}^{2,ZC}$and $IEP_{12m}^{2,ZC}$ got 20bp monthly contribution. The surprising results follow from Table 1, which concludes the spot return contributions for different strategies. In line with the literature, the carry portfolio earns -7bp per month, but the 6m and 12m zero-cost implied eigenportfolios are revealed to contribute shocking 22bp and 28bp spot returns, respectively. So the 6m and 12m option market implied allocation does not only offer good carry returns, but it also seems to be able to select the allocation combination with expected positive spot contributions.

Third, we separete the option implied extra leverage between the short and long leg, $Orth_\tau = IEP_\tau^{2,ZC} - IEP_\tau^2$, and examine wether these market implied "orthogonality factors" contain any usefull information. The leverage of the IEP_τ^2 portfolios varies between -80% and 80% avaraging around zero. SSD tests show $Orth_\tau$ highly statistically significant (1% and 5% level) dominance over HML_{FX}. This is not surprising, since the Table 5 reports 20bp to 36bp expected mothly return, with positive skewness with roughly half of the volatility compared

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to carry factor. So the reversal of this orthogonality factor seems to be the component mainly driven by positive expected spot prices.

Finally, we analyze the information contents of 6m-1m VRP term structure of the option implied dollar factor (IEP^1_τ). As reported in table XX, the explanatory power of the 6m-1m VRP slope on the risk factor controlled eigenportfolio retuns had a statistically significant β on IEP^2_{6m}, IEP^2_{12m}, all four $IEP^{2,ZC}_\tau$ and for all four $Orth_\tau$. Their explanatory power varied from 2.6% to surprisingly large 12%, where as the β varied between 0.4 and 0.96. All the VTS timed portfolios turned out to SSD dominate the carry strategy in either 10% or 5% level.

Table 9. We follow the stochastic dominance test setup of Linton et al. (2005) and define that the SSD and TSD between the benchmark HML_{FX} and an alternative strategy, by comparing functions of the cumulative return distributions $F_k(x)$ of the different investment strategies k. .This table represents the second (SSD) and third (TSD)order stochastic dominance test results for 21 alternative carry strategies against the benchmark HML_{FX}

	FSD		SSD		TSD	
	$H^1_0 : HML_{FX} \succ_s z$	$H^2_0 : z \succ_s HML_{FX}$	$H^1_0 : HML_{FX} \succ_s z$	$H^2_0 : z \succ_s HML_{FX}$	$H^1_0 : HML_{FX} \succ_s z$	$H^2_0 : z \succ_s HML_{FX}$
IEP^2_{1m}	0.49	0.32	0.30	0.40	0.29	0.58
IEP^2_{3m}	0.50	0.18	0.19	0.49	0.26	0.50
IEP^2_{6m}	0.45	0.36	0.16	0.51	0.24	0.35
IEP^2_{12m}	0.23	0.40	0.13	0.40	0.20	0.27
$IEP^{2,ZC}_{1m}$	0.23	0.72	0.25	0.53	0.29	0.44
$IEP^{2,ZC}_{3m}$	0.36	0.73	0.26	0.62	0.23	0.46
$IEP^{2,ZC}_{6m}$	0.04**	0.53	0.22	0.48	0.22	0.31
$IEP^{2,ZC}_{12m}$	0.03**	0.79	0.16	0.23	0.16	0.10
$Orth_{1m}$	0.00	0.02	0.01**	0.51	0.06*	0.38
$Orth_{3m}$	0.01	0.00	0.001***	0.56	0.08*	0.37
$Orth_{6m}$	0.01	0.00	0.001***	0.56	0.07*	0.35
$Orth_{12m}$	0.04	0.00	0.001***	0.56	0.08*	0.35
$IEP^{2,ZC,VTS}_{1m}$	0.49	0.67	0.07*	0.74	0.09*	0.68
$IEP^{2,ZC,VTS}_{3m}$	0.10	0.71	0.06*	0.70	0.05*	0.63
$IEP^{2,ZC,VTS}_{6m}$	0.38	0.77	0.06*	0.70	0.04**	0.63
$IEP^{2,ZC,VTS}_{12m}$	0.23	0.81	0.03**	0.59	0.02**	0.50
HML^{VTS}_{FX}	0.12	0.67	0.03**	0.55	0.03**	0.52
$Orth^{VTS}_{1m}$	0.03	0.04	0.01**	0.62	0.06*	0.44
$Orth^{VTS}_{3m}$	0.02**	0.14	0.003***	0.62	0.04**	0.33
$Orth^{VTS}_{6m}$	0.00	0.04	0.002***	0.59	0.03**	0.27
$Orth^{VTS}_{12m}$	0.00	0.02	0.002***	0.56	0.02**	0.19

6 Conclusions

This paper provides a novel way of revealing the option embedded currency risk factor allocations for option implied RX and HML_{FX} in G10 currencies, covering the period 09/2008-08/2016. These forward looking market implied eigenportfolios create highly profitable carry strategies, $IEP^{2,ZC}_\tau$, that stochastically dominate HML_{FX} and earn both high carry and surprisingly high spot price contribution as well. We also find new empirical evidence that the

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expected carry returns of both $IEP_{\tau}^{2,ZC}$ and HML_{FX} are strongly related to the changes of the 6m-1m slope of variance risk premium term structure (VTS) of the implied dollar factor, IEP_{τ}^{1}.

First, we analyse the option implied zero-cost carry factors $IEP_{\tau}^{2,ZC}$. The first order Stochastic Dominance tests report very strong results for the superiority of $IEP_{6m}^{2,ZC}$ and $IEP_{12m}^{2,ZC}$ over the HML_{FX} strategy, where as in the 3m and 1m tenors, neither 1st, 2nd nor 3rd order are able to detect any dominance order between them. Also the traditional Sharpe ratio tests confirm the results, assigning statistically significant Sharpe ratios for $IEP_{6m}^{2,ZC}$ and $IEP_{12m}^{2,ZC}$.

When decomposing the traditional carry trade profitability, it is widely known to be driven solely by the positive interest rate differential, while the spot price tends not to have any contribution on the returns in the long run. In our sample the rate differential contributed HML_{FX} 24bp per month, whereas $IEP_{6m}^{2,ZC}$ and $IEP_{12m}^{2,ZC}$ earned 20bp monthly contribution. More interestingly, the surprising results follow from the spot return contributions of the different strategies. In line with the literature, the HML_{FX} portfolio earns -7bp per month spot contribution, but the 6m and 12m zero-cost implied eigenportfolios are revealed to contribute shocking respective 22bp and 28bp monthly spot returns. I.e. the 6m and 12m option market implied allocation does not only offer comparable carry contribution, but it also seems to be able to select the allocation combination with expected positive spot returns.

Second, even though the IEP_{τ}^{2} profitability seems to be inferior with respect to their traditional counterparty, HML_{FX}, the stochastic dominance tests are unable to indicate any statistically significant dominance structure at any (SD) order between any of the four implied eigenportfolio and HML_{FX}. Benefits of these option implied risk factors seem to exist in the diversification front. As the RX and HML_{FX} risk factors experience stunning 60% correlation over the whole observation period among the G10 currencies, the respective realized correlation between IEP_{τ}^{2} and IEP_{τ}^{1} varies only between 10% and 21%. The large RX exposure seems to be driving the carry profits of HML_{FX}, and deteriorate the diversification effect. Constructed to be orthogonal with respect to IEP_{τ}^{1}, option implied carry factors seem to maintain their

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independence fairly well.

Finally, we analyze the information contents of 6m-1m VRP term structure slope of the option implied dollar factor (IEP^1_τ). The explanatory power of the 6m-1m VRP slope on the risk factor controlled eigenportfolio retuns had a statistically significant β on IEP^2_{6m}, IEP^2_{12m}, all four $IEP^{2,ZC}_\tau$, for all four $Orth_\tau$ and for HML_{FX}. Their explanatory power varied from 2.6% to surprisingly large 12%, whereas the β varied between 0.4 and 0.96. All the VTS timed portfolios turned out to SSD dominate the carry strategy in either 10% or 5% level.

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Table 10. This table presents the realized correlations for the new and traditional risk factors, during the entire sample.

	IEP^1_{1m}	IEP^1_{3m}	IEP^1_{6m}	IEP^1_{12m}	IEP^2_{1m}	IEP^2_{3m}	IEP^2_{6m}	IEP^2_{12m}	$IEP^2_{1m,zc}$	$IEP^2_{3m,zc}$	$IEP^2_{6m,zc}$	$IEP^2_{12m,zc}$	RX	HML_{FX}
IEP^1_{1m}	1.000	1.000	0.999	0.999	0.181	0.106	0.169	0.212	0.413	0.487	0.521	0.549	0.992	0.654
IEP^1_{3m}		1.000	1.000	0.999	0.183	0.101	0.166	0.209	0.419	0.485	0.520	0.547	0.994	0.653
IEP^1_{6m}			1.000	1.000	0.176	0.096	0.158	0.201	0.412	0.478	0.511	0.539	0.993	0.648
IEP^1_{12m}				1.000	0.173	0.093	0.153	0.196	0.408	0.473	0.505	0.534	0.991	0.647
IEP^2_{1m}					1.000	0.860	0.854	0.788	0.877	0.774	0.765	0.709	0.151	0.570
IEP^2_{3m}						1.000	0.962	0.881	0.765	0.851	0.814	0.736	0.050	0.564
IEP^2_{6m}							1.000	0.928	0.780	0.845	0.864	0.797	0.118	0.644
IEP^2_{12m}								1.000	0.727	0.788	0.810	0.872	0.155	0.685
$IEP^2_{1m,zc}$									1.000	0.877	0.875	0.814	0.387	0.685
$IEP^2_{3m,zc}$										1.000	0.976	0.905	0.438	0.741
$IEP^2_{6m,zc}$											1.000	0.936	0.476	0.788
$IEP^2_{12m,zc}$												1.000	0.499	0.813
RX													1.000	0.601
HML_{FX}														1.000

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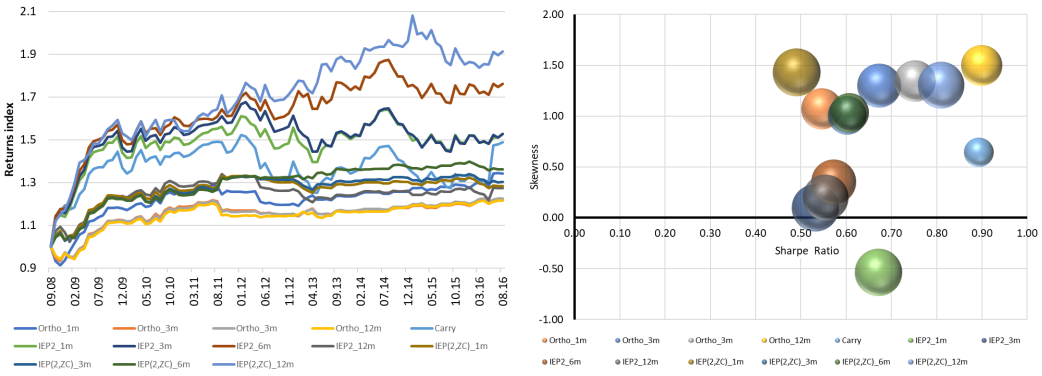
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Table 11. This table presents the descriptive statistics for all 12 eigenportfolios. These are monthly returns and the time sereis spans from 09/2008-08/2016.

	IEP^{1}_{1m}	IEP^{2}_{1m}	IEP^{1}_{3m}	IEP^{2}_{3m}	IEP^{1}_{6m}	IEP^{2}_{6m}	IEP^{1}_{12m}	IEP^{2}_{12m}
Raw return Mean	-0.045	0.065	-0.037	0.127	-0.034	0.206	-0.032	0.273
Max (in %)	7.81	7.66	7.93	5.03	7.91	5.47	7.89	5.84
Min (in %)	-11.06	-6.52	-10.90	-6.80	-10.49	-7.70	-10.32	-8.80
Std. Dev.	3.06	2.25	3.06	2.03	3.06	2.01	3.08	2.03
Skewness	-0.42	0.31	-0.40	-0.01	-0.37	-0.30	-0.36	-0.56
Kurtosis	1.55	1.29	1.50	0.80	1.34	1.81	1.26	3.48
Sharpe Ratio	-0.01	0.03	-0.01	0.06	-0.01	0.10	-0.01	0.13

	$IEP^{1,ZC}_{1m}$	$IEP^{2,ZC}_{1m}$	$IEP^{1,ZC}_{3m}$	$IEP^{2,ZC}_{3m}$	$IEP^{1,ZC}_{6m}$	$IEP^{2,ZC}_{6m}$	$IEP^{1,ZC}_{12m}$	$IEP^{2,ZC}_{12m}$
Raw return Mean	-0.045	0.375	-0.037	0.337	-0.034	0.420	-0.032	0.478
Max (in %)	7.81	12.21	7.93	5.51	7.91	5.50	7.89	5.98
Min (in %)	-11.06	-13.59	-10.90	-12.15	-10.49	-12.15	-10.32	-13.22
Std. Dev.	3.06	2.96	3.06	2.58	3.06	2.54	3.08	2.57
Skewness	-0.42	-0.44	-0.40	-0.95	-0.37	-1.09	-0.36	-1.36
Kurtosis	1.55	6.65	1.50	4.77	1.34	5.26	1.26	7.43
Sharpe Ratio	-0.01	0.13	-0.01	0.13	-0.01	0.16	-0.01	0.19



(a) Variance Term Structure sorted portfolio returns

(b) This graph illustrates the Sharpe ratio(x-axis), skewness(y-axis) and kurtosis(ball size) of the Variance term structure sorted strategies

Figure 6. my caption

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Table 12. The predictive power of the first eigen vector variance risk premium slope for the implied eigen vecror returns. This table reports the estimated coeffcients for the following regressions: $rx_{i,t+1} = \beta_1 + \beta_2(VRP^1_{t,6m} - VRP^1_{t,1m})$,where $rx_{i,t+1}$ is the excess return on 12 implied eigenportfolio retuns, and the slope of variance risk premium for zero-cost first implied eigenportfolio(IEP^1), the difference between 6 month and 1 month VRP's at $t-1$. The standard errors are corrected by Newey-West with 12 lags and the standard deviations are reported in parenthesis. and *** represent the usual 10, 5 and 1% significance levels. The sample period runs from Sep 2008 to Aug 2016.Slope VRP in the table stands for $VRP(IEP^1_{6m}) - VRP(IEP^1_{1m})$ and $Ortho_\tau$.

	IEP^1_{1m}	IEP^2_{1m}	IEP^1_{3m}	IEP^2_{3m}	IEP^1_{6m}	IEP^2_{6m}	IEP^1_{12m}	IEP^2_{12m}
Slope VRP	0.709	0.325	0.708	0.340	0.667	0.402	0.649	0.516
	(2.18)**	(1.230)	(2.17)**	(1.561)	(2.04)**	(1.87)*	(1.973)*	(2.40)**
constant	-0.0003	0.0002	-0.0002	0.0014	-0.0002	0.0022	-0.0001	0.0020
	(-0.008)	(0.009)	(-0.06)	(0.661)	(-0.049)	(1.06)	(-0.004)	(1.41)
$R\hat{2}$	0.04	0.01	0.04	0.02	0.03	0.26	0.03	0.05

	$IEP^{1,ZC}_{1m}$	$IEP^{2,ZC}_{1m}$	$IEP^{1,ZC}_{3m}$	$IEP^{2,ZC}_{3m}$	$IEP^{1,ZC}_{6m}$	$IEP^{2,ZC}_{6m}$	$IEP^{1,ZC}_{12m}$	$IEP^{2,ZC}_{12m}$
Slope VRP	0.709	0.703	0.708	0.806	0.667	0.860	0.649	0.955
	(2.18)**	(2.23)**	(2.17)**	(2.996)***	(2.04)**	(3.273)***	(1.973)*	(3.638)***
constant	-0.0003	0.0040	-0.0002	0.0036	-0.0002	0.0044	-0.0001	0.0050
	(-0.008)	(1.325)	(-0.06)	(1,41)	(-0.049)	(1.786)*	(-0.004)	(2.030)**
$R\hat{2}$	0.04	0.04	0.04	0.08	0.03	0.09	0.03	0.12

Table 13. This table presents the descriptive statistics for the first eigenportfolio variance risk premiums, $VRP(IEP^{1,zc})$, calculated as $VRP_{t,\tau} = \mathbb{E}^{\mathbb{P}}(RV_{t,\tau}) - \mathbb{E}^{\mathbb{Q}}(RV_{t,\tau})$

	$VRP(IEP^{1,zc}_{1m})$	$VRP(IEP^{1,zc}_{3m})$	$VRP(IEP^{1,zc}_{6m})$	$VRP(IEP^{1,zc}_{12m})$	$VRP(IEP^{1,zc}_{6m-1m}))$
Mean	-0.0001	-0.0002	-0.0003	-0.0004	-0.0002
Std	0.0066	0.0050	0.0056	0.0068	0.0095
Skew	4.25	2.70	1.13	-0.72	-3.85
Kurt	27.37	9.97	6.78	7.05	29.62
Sharpe	-0.0147	-0.0488	-0.0597	-0.0568	-0.0253

Table 14. This table presents the descriptive statistics for the eight eigenportfolio IEP^2_τ and HML_{FX} strategies, when the term structure of variance risk premium for IEP^1_τ, $TSVRP_t = VRP^{IEP1}_{t,6m} - VRP^{IEP1}_{t,1m}$, is used to time the long/short position in the strategies. Positive steepness of the TSVRP is used as signal to go long and negative TSVRP suggests short position in strategies. Our data spans from 09/2008 to 08/2016.

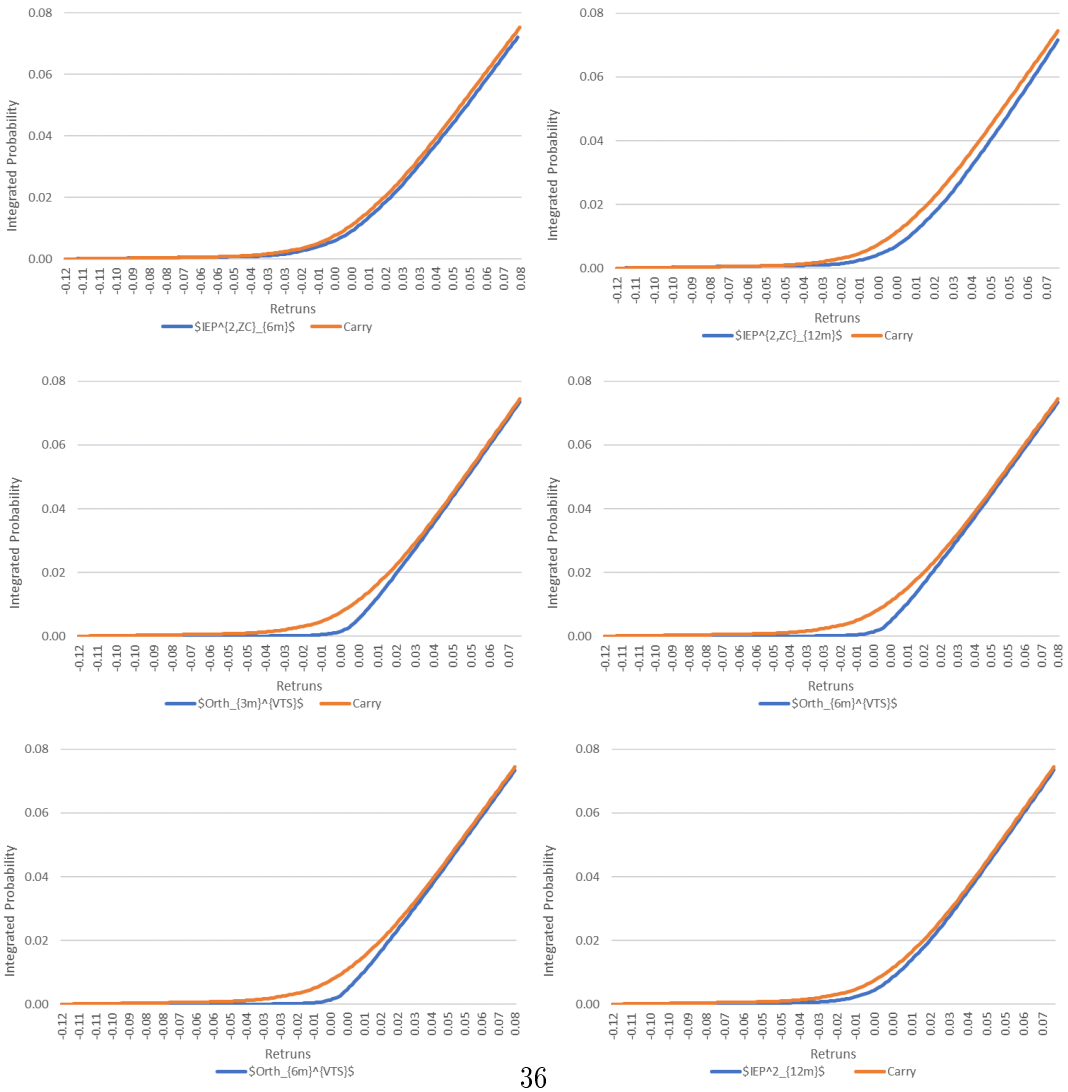
	$IEP^{2,VTS}_{1m}$	$IEP^{2,VTS}_{3m}$	$IEP^{2,VTS}_{6m}$	$IEP^{2,VTS}_{12m}$	$IEP^{2,ZC,VTS}_{1m}$	$IEP^{2,ZC,VTS}_{3m}$	$IEP^{2,ZC,VTS}_{6m}$	$IEP^{2,ZC,VTS}_{12m}$	Car^{VTS}_{1m}
Mean	0.0031	0.0021	0.0021	0.0020	0.0042	0.0045	0.0045	0.0060	0.0068
Std	0.0160	0.0137	0.0129	0.0128	0.0296	0.0257	0.0254	0.0255	0.0265
Skewness	-0.536	0.097	0.353	0.197	1.431	1.008	1.034	1.303	0.643
Kurtosis	5.485	5.556	4.965	5.179	5.623	3.633	3.790	5.363	2.125
Sharpe	0.672	0.532	0.573	0.555	0.491	0.602	0.607	0.810	0.893

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Table 15. This table presents the descriptive statistics for Price of Orthogonality strategies, when the term structure of variance risk premium for IEP_{τ}^{1}, $TSVRP_t = VRP_{t,6m}^{IEP1} - VRP_{t,1m}^{IEP1}$, is used to time the long/short position in the strategies. Positive steepness of the TSVRP is used as signal to go long and negative TSVRP suggests short position in strategies. Our data spans from 09/2008 to 08/2016.

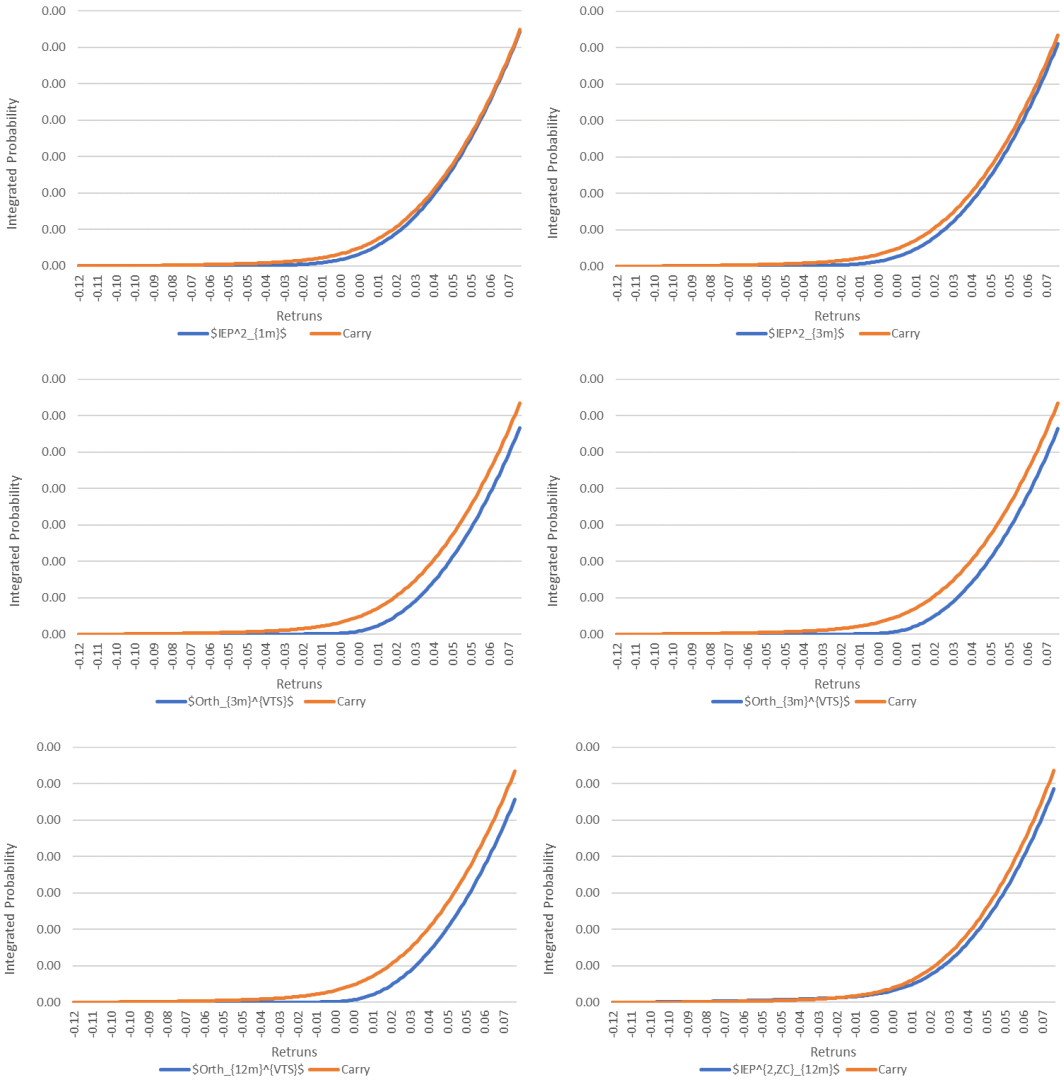
	$Ortho_{1m}^{VTS}$	$Ortho_{3m}^{VTS}$	$Ortho_{6m}^{VTS}$	$Ortho_{12m}^{VTS}$	Car_{1m}^{VTS}
Mean	0.0025	0.0026	0.0028	0.0033	0.0068
Std	0.0161	0.0136	0.0128	0.0125	0.0265
Skewness	1.072	1.301	1.345	1.498	0.643
Kurtosis	4.221	4.722	4.195	4.135	2.125
Sharpe	0.546	0.673	0.752	0.899	0.893

Figure 7. This table presents the second order stochastic dominance (SSD) of selected strategies against the benchmark HML_{FX}. Our data spans from 09/2008 to 08/2016.



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Figure 8. This table presents the third order stochastic dominance (TSD) of selected strategies against the benchmark HML_{FX}. Our data spans from 09/2008 to 08/2016.



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